

KAARINA YLTOINEN, Finland

WMO: 028280

Lat: 60.387N

Lon: 22.551E

Elev: 17

StdP: 14.69

Time Zone: 2.00 (EUE)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -5.4	(c) 0.4	(d) -9.6	(e) 3.3	(f) -4.6	(g) -3.9	(h) 4.5	(i) 1.5	(j) 14.5	(k) 21.9	(l) 12.8	(m) 22.6	(n) 2.6	(o) 30	(p) N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.2	80.9	67.0	77.1	65.6	73.8	63.3	70.0	77.0	67.5	74.5	65.4	71.5	4.3	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
67.4	100.9	73.4	65.2	93.3	70.4	63.0	86.1	68.2	33.9	77.2	32.0	74.5	30.3	71.4	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 12.0	(b) 10.1	(c) 9.0	(e) -9.1	(f) 84.2	(g) 8.9	(h) 4.7	(i) -15.5	(j) 87.5	(k) -20.7	(l) 90.3	(m) -25.7	(n) 92.9	(o) -32.1	(p) 96.3		
(4) 12.0	(5) 10.1	(6) 9.0	(7) -9.4	(8) 71.6	(9) 8.9	(10) 3.3	(11) -15.8	(12) 73.9	(13) -21.0	(14) 75.8	(15) -26.0	(16) 77.6	(17) -32.4	(18) 80.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	42.6	22.5	24.9	29.7	39.3	50.5	57.7	63.3	60.7	53.2	42.3	36.3	29.4	(6)
(7)		DBStd	15.60	10.54	10.28	7.80	5.62	7.38	5.64	5.35	4.82	5.47	6.36	7.40	9.25	(7)
(8)		HDD50	3923	852	704	628	325	83	6	0	0	31	246	411	637	(8)
(9)		HDD65	8257	1317	1124	1093	772	451	230	98	150	355	704	861	1102	(9)
(10)		CDD50	1215	0	0	0	3	99	236	412	332	126	7	0	0	(10)
(11)		CDD65	73	0	0	0	0	2	10	45	16	0	0	0	0	(11)
(12)		CDH74	723	0	0	0	0	47	127	434	114	1	0	0	0	(12)
(13)		CDH80	135	0	0	0	0	3	15	96	21	0	0	0	0	(13)
(14)	Wind	WSAvg	3.8	3.8	3.6	3.8	3.6	4.0	3.6	3.7	3.4	3.6	3.9	4.3	4.4	(14)
(15)	Precipitation	PrecAvg	26.7	2.3	1.6	1.3	1.3	1.3	2.1	2.7	3.3	2.3	3.0	2.7	2.7	(15)
(16)		PrecMax	35.0	4.2	3.7	3.5	2.2	2.8	3.7	5.1	7.1	5.8	7.2	6.2	5.9	(16)
(17)		PrecMin	18.9	0.2	0.0	0.0	0.2	0.2	0.4	0.0	0.9	0.4	0.6	0.3	0.2	(17)
(18)		PrecStd	3.7	1.1	0.9	0.7	0.6	0.7	0.8	1.6	1.4	1.3	1.6	1.3	1.3	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	40.4	42.8	50.1	65.6	79.0	82.1	86.7	83.0	71.0	58.4	50.8	47.3	(19)
MCWB			38.8	37.9	41.8	50.7	63.0	66.3	68.8	69.1	62.5	53.9	49.3	45.6	(20)	
2%		DB	38.3	39.4	45.6	58.9	74.1	77.8	82.5	76.7	67.3	55.3	48.6	44.0	(21)	
		MCWB	37.1	36.7	39.3	47.6	59.2	64.0	68.9	66.7	60.4	52.5	47.7	42.5	(22)	
5%		DB	36.3	37.0	41.8	54.3	70.2	73.4	78.8	73.3	64.8	53.7	46.7	41.8	(23)	
		MCWB	35.4	36.1	37.2	45.4	57.8	61.0	67.9	64.3	59.1	51.8	45.8	40.6	(24)	
10%		DB	34.7	35.6	38.8	50.4	65.5	69.8	75.3	70.5	62.3	52.0	45.1	39.9	(25)	
		MCWB	33.9	34.9	35.5	43.1	55.0	59.2	65.8	62.6	58.3	50.4	44.1	39.0	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	39.2	39.5	43.0	51.9	64.7	69.1	73.5	71.2	64.0	55.8	49.8	46.3	(27)
MCDWB			39.9	41.9	47.9	63.6	72.9	77.7	81.4	77.9	68.6	56.9	50.3	46.9	(28)	
2%		WB	37.4	37.4	40.1	48.6	61.8	66.2	71.4	68.3	62.2	53.8	48.0	42.9	(29)	
		MCDWB	38.1	38.4	44.5	56.6	71.3	75.2	78.5	74.7	65.6	54.5	48.6	43.6	(30)	
5%		WB	35.7	36.2	38.0	46.3	59.0	63.5	69.0	66.1	60.6	52.4	46.1	40.8	(31)	
		MCDWB	36.3	37.0	40.7	53.2	68.7	71.1	75.9	70.7	63.5	53.1	46.6	41.5	(32)	
10%		WB	34.1	34.9	36.5	43.9	55.7	60.9	66.7	64.4	59.0	50.6	44.4	39.0	(33)	
		MCDWB	34.7	35.6	38.4	49.3	64.5	67.9	73.1	68.9	61.5	51.8	45.0	39.7	(34)	
(35)	Mean Daily Temperature Range	5% DB	MDBR	10.6	10.7	13.9	16.9	19.4	17.4	18.2	17.0	15.1	12.1	7.5	8.4	(35)
MCDBR			6.8	10.5	16.0	25.7	27.0	24.8	23.8	21.7	18.5	13.4	7.8	9.0	(36)	
MCWBR			6.4	9.0	10.8	15.5	14.6	13.2	12.0	12.5	12.5	10.6	7.3	8.3	(37)	
5% WB		MCDBR	6.5	9.4	13.7	22.8	25.4	21.7	20.2	18.0	15.7	10.8	7.7	8.6	(38)	
		MCWBR	6.3	8.2	10.1	14.3	14.7	12.6	11.6	11.3	11.5	9.7	7.5	8.3	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.276	0.313	0.334	0.373	0.368	0.361	0.389	0.379	0.350	0.333	0.303	0.386	(40)	
taud		2.149	2.228	2.308	2.322	2.401	2.448	2.392	2.413	2.483	2.397	2.213	2.613	(41)		
Ebn at Noon		149	213	250	261	272	276	264	257	246	207	138	103	(42)		
Edh at Noon		13	22	28	33	33	32	33	30	24	19	13	9	(43)		
(44)	All-Sky Solar Radiation	RadAvg	77	257	732	1229	1679	1760	1657	1304	801	349	97	47	(44)	
(45)		RadStd	14	51	89	124	165	139	162	137	101	56	18	5	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (58 neighbors)		+0.74	N/A	N/A	N/A	N/A	N/A	-239	N/A	N/A	N/A

Nomenclature: See separate page